INCH-POUND

MIL- M-83805 (84)

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MILITARY SPECIFICATION

MIXER, CONCRETE, TRUCK MOUNTED, 8 CUBIC YARD CAPACITY, 6 X 4 OR 6 X 6

This specification is approved for use within WR-ALC, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers concrete mixing, agitating, and distributing trucks of conventional design, either front or rear discharge. Units are 6 X 4 or 6 X 6 and shall be capable of extended off-road operation.

1.2 Classification. Either a six wheel drive or four wheel drive unit will be specified in the ordering data (see 6.2). Both front and rear discharge units may be offered in response to this specification and will compete equally.

2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

STANDARDS

Federal

FED-STD-595	Colors.

Military

MIL-STD-209	Slinging and Tie Down Provisions For Lifting and Tying Down of Military Equipment.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: WR-ALC/MMVRG, Robins AFB, GA 31098-5609 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 3895

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MIL-STD-831	Test Reports, Preparation of.
MIL-STD-1791	Designing for Internal Aerial Delivery in Fixed Wing Aircraft.

DESIGN HANDBOOKS

Air Force Systems Command (AFSC)

AFSC DH 1-11	Air Transportability

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099).

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DEPARTMENT OF TRANSPORTATION (DoT)

Federal Motor Carrier Safety Regulations (FMCSR).
Federal Motor Vehicle Safety Standards (FMVSS).

(Applications for copies should be addressed to The Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)

Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines.
Interstate Motor Carrier Noise Emission Standards.

(Applications for copies should be addressed to the Public Affairs Office, Environmental Protection Agency, Rockville, MD 20852.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

THE TIRE AND RIM ASSOCIATION, INC. (TRA)

TRA Yearbook

(Application for copies should be addressed to the Tire and Rim Association Inc., 3200 West Market Street, Akron OH 444313)

THE SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE J534	Lubrication Fittings.
SAE J688	Truck Ability Prediction Procedures.
SAE J899	Operator's Seat Dimensions for Off-road Self-propelled Work Machines.
SAE J925	Minimum Access Dimensions for Construction and Industrial Machinery.

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(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Dr. Warrendale PA 15096)

TRUCK MIXER MANUFACTURER'S BUREAU (TMMB)

Truck Mixer and Agitator Standards.

(Application for copies should be addressed to the Truck Mixer Bureau, 900 Spring Street, Silver Spring, MD 20910.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS.

3.1 Description. The unit shall be a commercial truck transit mixer for mixing, transporting, and dispensing concrete. The mixer shall consist essentially of a frame, rotating drum, charging hopper, discharge chute, and watering system. The mixer shall be mounted on a diesel engine driven truck. As used herein, the term mixer refers to the mixing unit only. The term truck mixer refers to both the truck and the mixer.

3.1.1 Commerciality. The truck mixer shall be the latest model of the standard product of the contractor and shall have demonstrated industry acceptability by having been manufactured, marketed, and sold in significant numbers to the commercial public for the same or similar application, except as specified herein, for at least one year prior to invitation for bids or request for proposal. Product improvements incorporated in the commercial production unit during this 1 year period are acceptable. When specific requirements are not stated herein, all items listed as standard equipment in the contractor's published specification brochures and catalogs, or normally furnished to commercial customers as standard equipment, shall be furnished. The truck mixer shall be equipped with optional equipment as specified herein. Optional equipment is defined as equipment not standard with the truck mixer but which has been furnished to the commercial customer in the past, such as special features and allied equipment.

3.1.2 Integrity. Each unit furnished shall be complete and configured such that it is capable of providing its intended service upon arrival at its final destination.

a. Exceptions:

(1) When otherwise specified in section 5 and packaging instructions in the contract.
(2) When to do so would violate Federal, State or other applicable regulations, or create a potentially hazardous or damaging condition.

3.1.3 Operating conditions. Truck mixers must be capable of extended storage and operation in any and all of the following environments and conditions.

a. Temperatures- 110°F ambient air with full impact of solar radiation for both storage and operation. -40°F cold storage environment. -20° F cold operation temperature. 0° cold starting temperature. If winterization is specified (see 6.2), cold temperatures are as specified in 6.2. Cold temperatures are that of ambient air without the benefit of solar radiation.

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b. Truck mixer shall not be damaged or any function impaired by prolonged exposure to wind blown sand, dust, or precipitation.
c. Truck mixer shall withstand prolonged exposure to corrosive coastal environments.
d. Truck mixer shall withstand extended off-road operation such as is required when being supplied by a remote crushing plant and when delivering to a remote location.
e. Truck Mixer shall be capable of extended operation at 10,000 feet altitude.

3.1.4 Life. The truck mixer shall be designed and constructed to provide a minimum of twelve years service. The ability to cost effectively maintain a safe and efficient vehicle constitutes a serviceable vehicle.

3.2 Maintainability. All maintenance described in this specification shall be capable of being performed by one mechanic using common hand tools. Design and construction of the truck mixer shall permit ready accessibility to all parts for maintenance and repair in the field. Access shall be in accordance with SAE J925 minimum access dimensions. Replacement and adjustment of component parts and accessories shall be accomplished with minimum drainage of fluids and disturbance to other parts.

3.2.1 Filling, draining, and checking provisions. All fluid systems shall possess a convenient means of filling, draining, and checking the level of the system. All drains shall conduct fluid directly to the ground in a stream with out spilling onto other components. Engine, transmission, and hydraulics shall have permanently magnetized drain plugs. Fluid level indicators shall identify the type of fluid, state whether the level should be checked cold, hot, or with engine running, and identify appropriate add and full levels.

3.2.2 Lubrication points. Lubrication points shall be provided for all surfaces requiring lubrication. Pressure relief fittings shall be supplied where the use of pressure lubrication equipment may damage seals. Lubrication fittings shall be in accordance with SAE J534 and be located so as to be clearly visible and easily serviced with a grease gun with a 10 inch flexible hose.

3.2.3 Positioning. All parts which could cause improper performance if positioned incorrectly shall possess asymmetrical mounting devices which prevent incorrect positioning.

3.2.4 Maintenance operations list. Any four or more operations shall be able to be performed by one person in the sum of time allotted for each operation.

3.2.5 Materials. The components, pieces and parts incorporated in the mix truck may be newly fabricated from recovered materials to the maximum extent practical, provided the mix trucks produced meet all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the mix truck. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials.

a. Remove, replace, and adjust all engine driven belts - 1 hour.
b. Remove and replace voltage or current regulator - 15 min.
c. Remove and replace generator and adjust generator belts - 30 min.
d. Remove and replace all filter elements, screens, and strainers on the hydraulic system - 30 min.
e. Drain cooling system, remove and replace engine cooling hoses, and refill cooling system - 1 hour.
f. Drain engine lubricating oil, remove and replace oil filter element, and refill crankcase - 30 min.
g. Remove and replace fuel filter elements - 30 min.
h. Remove and replace transmission fluid filter elements and strainer - 45 min.
i. Drain and refill transmission and torque converter - 20 min.
k. Disconnect battery leads, remove and replace batteries, and reconnect battery leads - 15 min.

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3.3 Reliability. The contractor shall furnish a unit capable of providing the services specified in the contract under the specified conditions for the specified life. No unit shall be furnished which possesses parts or qualities known to or suspected of causing premature failure of the vehicle. Information regarding such parts or qualities shall be forwarded to the contracting officer as it is recognized throughout the contracting period and specified life of the vehicle. It is the contractors responsibility to correct and, when possible, prevent any damage to the unit caused by latent defects in material or production of the units.

3.4 Safety. All exposed parts subject to high operational temperatures or electrically energized shall be located, insulated, enclosed, or guarded so as to prevent hazards to operating personnel. All moving parts which present a hazard shall be guarded.

3.4.1 Steps, grabrails, and platforms. Steps, grabrails, platforms, etc. shall be provided to ensure safe and convenient access to all control, maintenance, and inspection points. Steps, footholds, and walkways shall be expanded metal, perforated metal, or possess a durable nonskid surface.

3.4.2 Noise level. Noise level shall be in accordance with EPA Interstate Motor Carrier Noise Emission Standards.

3.4.3 Backup alarm. A backup warning alarm audible above the noise of the vehicle shall be installed.

3.4.4 Sharp edges. All nonfunctional edges shall be rounded or broken so as to not present a safety hazard.

3.5 Power train.

3.5.1 Engine. The truck mixer shall be powered by a standard commercial diesel engine. The horsepower, torque, and speed demanded of the engine during the performance of the truck mixer's intended function (see 6.1) or during the performance of any specified functions shall not exceed the engine's previous horsepower, speed, or torque rating for truck mixer application. Engine shall comply with EPA regulations governing Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines applicable on the date of manufacture. An engine governor shall be installed, set and sealed to limit engine speed to maximum recommended speed.

3.5.2 Transmission. Truck mixer shall have a torque converter, an automatic, full power shift transmission with at least 4 speed ranges, and a drop box with at least two speeds. Torque converter, transmission, and drop box shall be capable of withstanding all loads and speeds normally encountered in the truck mixer's operation. Torque converter shall feature automatic lockup in at least the top two ranges.

3.5.3 Differentials. On 6 X 4 units, differentials shall be one of the following types-

a. Standard interaxle and interwheel differentials.
b. Manual or Automatic locking or limited slip interwheel differentials and standard interaxle differential. Manual controls to be located in cab if applicable.

On 6 X 6 units, differentials shall be one of the following types-

a. Standard interaxle and interwheel differentials. Front wheel drive shall be engageable and disengageable from the cab while the truck mixer is rolling or unit shall be full time all wheel drive.
b. Manual or automatic locking or limited slip interwheel differential(s) on rear drive axles and an interaxle differential. Manual controls to be located in cab if applicable. Front wheel drive shall be engageable and disengageable from the cab while the truck is rolling, or, the unit shall be full time all wheel drive.

3.6 Mixer.

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3.6.1 Drum. The drum shall be the open end type with an opening end diameter of not less than 34 inches. Capacity and volume of the drum shall conform to table 1 of the TMMB Truck Mixer and Agitator Standards for an 8 cubic yard mixer. The basis for determining drum capacity and the formulas used for calculating drum volume shall be as specified in the TMMB Truck Mixer and Agitator Standards. Drums of larger nominal capacity will be acceptable provided maximum and minimum gross volumes conform to TMMB Truck Mixer and Agitator Standards. The drum shall be watertight and made of corrosion resistant steel of not less than 0.1875 inch nominal thickness on the mixing portion of the drum, with allowable variations at the head and tail sections which would provide an equivalent wear rate to that of the 0.1875 inch sections. All sections shall be of welded construction to form a smooth, uniform surface enabling the entire drum to withstand the maximum loads or stresses during heavy operations. The drum shall be equipped with an adequate number of blades to thoroughly mix the concrete and to quickly discharge the homogeneous batch. The blades shall be of the same material as the drum with same thickness or a material with an equivalent wear rate. When open, the hatch(s) shall provide a clear opening having a minimum diameter of not less than 20 inches for round hatches and not less than 12.375 by 18.75 for other shaped hatches. When hatch(s) are closed, drum shall be watertight. Drum rollers shall be made of high strength steel, precision machined, and treated to resist wear. Drum roller track shall run true and flat on the rollers.

3.6.2 Water system. Mixer shall be equipped with an air-pressurized water system. The water tank(s) and water system shall comply with the requirements of the TMMB Truck Mixer and Agitator Standards, except the water tanks shall not be less than 240 gallons capacity. Sight gages shall be visible to the operator while standing at the external controls. Mixer shall be provided with a dual water injection system, a 25 foot wash off hose with adjustable spray and shutoff nozzle, and drain plugs to completely drain the system. An air blowout system shall be used to accomplish normal draining of the system. Plastic piping is not acceptable. The water meter shall be of the resetable counter type, with totalizer and automatic stop valve.

3.6.3 Charging hopper. Unless otherwise specified, the manufacturer shall furnish his standard commercial stationary or swing-out type charging hopper.

3.6.4 Discharge chute. The discharge chute shall consist of an upper and a lower section. The discharge end of the chute shall be horizontally and vertically adjustable. The upper section shall pivot to provide 180° rotation in the horizontal plane. Vertical movement shall be accomplished through hydraulic cylinder attached to the upper section. Upper and lower sections shall be hinged together allowing the lower section to fold over the upper section for stowage and transit. Extension chute section(s) shall be furnished which shall provide a total chute length of not less than 12 feet. Chute extension(s) shall be attachable in one minute and detachable in one minute by one person. Stowage shall be provided for securely stowing chute extension(s). Chute with all extensions attached shall have a smooth and unobstructed trough.

3.6.5 Drum power transmission. Unless otherwise specified (see 6.2), either an open or closed type hydraulic transmission circuit will be acceptable. The hydraulic pump shall be mounted on and driven by the truck's power takeoff and shall be the manufacturer's standard commercial unit. All hoses and fittings shall be capable of withstanding a maximum bursting pressure of at least four times the working pressure and a maximum proof pressure of at least two times the working pressure. Hoses shall be protected from chaffing through the use of supports and protective devices. The hydrostatic transmission shall be equipped with a full flow filter having a micron rating not greater than 25. The inlet side shall have a suction strainer or filter to prevent the entrance of foreign material. The hydraulic fluid reservoir shall possess a vent equipped with a filter having a micron rating not greater than 25. Hydraulic oil shall be of the non-corrosive type with the characteristics to perform through the temperature range of 10°F through 325°F. Reducer sprockets, shafts, and chains shall be heat treated steel. Chains shall be single strand, precision roller type having a minimum pitch of 1.75 inch (140 chain). A chain tensioner shall be provided capable of being properly set by one man in five minutes using common hand tools. An automatic oiler shall be provided to lubricate the chain. Direct gear drive is also acceptable; and when so provided, shall possess a steel or aluminum case with heat treated spur gears.

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3.6.6 Drum revolution counter. An electrically operated, limit switch, or mechanical type revolution counter shall be furnished.

3.6.7 Mixer controls. Truck mixer shall be equipped with two sets of mixing controls. Each set of controls shall control engagement and disengagement of the power train, rotary speed, and direction of rotation. One set of controls shall be located in the truck's cab; the second set shall be located on or near the left rear corner of the truck for rear discharge units or on or near the left front corner for front discharge units. Cab controls shall be located within easy reach of the operator when sitting in a normal driving position. The second control set shall be conveniently located for an operator standing on the ground. All controls shall be clearly identified as to their function with permanently affixed identification plates of corrosion resistant material.

3.7 Systems and components.

3.7.1 Truck chassis. The mixer shall be mounted on a commercial truck chassis having a minimum gross vehicle weight rating of not less than 62,000 pounds. Except as otherwise specified herein, all chassis items shall be as represented in the chassis manufacturer's technical data. Technical data shall be limited to specifications and technical material, identical to that furnished the authorized company representative for the selection of truck models and mixers, and shall be furnished to the engineering office of the procuring agency 30 days prior to first article testing or delivery, whichever comes first. Since the truck mixer to be procured is a commercial truck mixer, truck mixer shall be in compliance with all DoT Federal Motor Carrier Safety Regulations (FMCSR) and Federal Motor Vehicle Safety Standards applicable to nonmilitary, nonexport motor vehicles on the date of manufacture.

3.7.2 Electrical system. The truck mixer shall be equipped with a 12 volt starting and lighting system, with not less than a 60 amp generator or alternator. Batteries shall be maintenance free, with a total of at least 2000 cold cranking amps. All lights, reflectors and wiring shall be as specified herein and shall conform to FMCSR, sections 393.12, 393.13, 393.19, 393.20, 393.22 through 393.26(d), 393.27, through 393.31, 393.33.

3.7.3 Fuel system. Fuel system shall conform to FMCSR, section 393.65. Fuel tank(s) capacity shall be at least 75 gallons. When more than one fuel tank is furnished, means to assure equalized fuel level in each tank shall be provided. Fuel filters shall be installed at each tank and before the engine.

3.7.4 Cooling system. The truck shall be equipped with a heavy-duty cooling system capable of enabling the truck mixer to function as specified at 10,000 feet altitude or in 110°F environment.

3.7.5 Brakes. Brakes shall conform to FMCSR, sections 393.45 through 393.52, as applicable. Service brakes and parking brakes shall conform to Federal Motor Vehicle Safety Standard No. 121.

3.7.5.1 Service brakes. The truck shall be equipped with full air brakes. The braking system shall be equipped with all necessary components as follows:

a. Air compressor, unloader type, engine driven and lubricated, air or water cooled, and having a capacity of not less than 12 cubic feet per minute.
b. Air storage reservoir(s) of at least 3000 cubic inches capacity. Each tank shall be equipped with a drain at the lowest point. There shall be a safety and check valve between the compressor and last reservoir tank.
c. Foot control, suspended or treadle type.
d. Air control valves.
e. Air pressure gage.
f. Low air pressure warning device, visible and audible.
g. Alcohol evaporator kit or equivalent.
h. Automatic moisture ejector.

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3.7.5.2 Parking brake. The truck shall be equipped with a parking brake control valve. Parking brake shall conform to FMCSR 393.41.

3.8 First Article. When specified (see 6.3), a sample shall be subjected to first article inspection in accordance with 4.4.

3.9 Performance.

3.9.1 Mobility. High and low speed requirements shall be met with truck mixer loaded to specified GVW and calculated according to SAE J688. The truck mixer shall ascent a 0.5% grade at 44 mph. With drum filled to mixing capacity and operating at not less than 2 rpm and not more than 6 rpm, the truck mixer shall be capable of maintaining a minimum speed of 35 mph over a level improved road. The low speed requirement shall be calculated with engine operating at not less than 35 percent of governed speed and shall not exceed 3 mph. Truck mixer shall be capable of extended travel on unimproved roads under all operating conditions.

3.9.2 Mixing and agitating. Mixing and agitating speeds shall be in accordance with the TMMB Truck Mixer and Agitator Standards.

3.10 Cab. The truck shall be furnished with a full width cab not less than 60 inches measured between door panels. Cab shall provide seating for two people. Seats shall be upholstered with manufactures dark, extreme service material. Driver's seat shall be adjustable in accordance with the recommended dimensions stated in SAE 899 for US 5th percentile female through 95th percentile male. Handles and steps shall be provided as necessary to ensure easy entry from both sides of the cab. Dual windshield washers and multispeed wipers shall be installed. A hot water cab heater with blower and fresh air intake shall be installed. Heater shall have floor and defroster vents. Instruments and controls shall be identified as to their function and installed to facilitate easy removal and servicing. The following equipment shall be furnished.

a. Ammeter or charging indicator.
b. Fuel gage.
c. Oil pressure gage or indicator.
d. Engine temperature gage or indicator.
e. Speedometer with recording odometer.
f. Ash receptacle.
g. Dual visors.
h. Tachometer.
i. Armrests.
j. Cab ventilator other than window.
k. Horn.
l. Interior lights.
m. Two seat belts with shoulder harnesses.
n. Outside rearview mirrors, mounted on each side of cab, each with not less than 85 square inches of reflective surface, and each with not less than three supporting arms.
o. All other equipment usually furnished as standard by the manufacturer on his commercial vehicles.

3.11 Additional equipment.

3.11.1 Bumper and two hooks. A channel type front bumper shall be furnished. Not less than two hooks or loops, or a tow tube for towing the truck mixer shall be furnished.

3.11.2 Power steering. Truck mixer shall be equipped with power steering assist. This system shall not prevent manual steering capability in the event of engine failure.

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3.11.3 Splash and stone throw protection. Rear wheels shall posses a metal guard above them of sufficient size and suitable located so as to prevent stones and mud from being thrown onto the rest of the truck mixer. Rubber mud flaps on the rear are acceptable. Front tires shall be located or guarded so as to prevent stones and mud from being slung onto the cab.

3.11.4 Tires. Tires shall be in accordance with the Tire and Rim Association Handbook's recommendations. Tires shall have mud and snow tread type.

3.12 Lifting provisions. Lifting provisions shall be in accordance with MIL-STD-209, types two or three.

3.13 Tie down provisions. Tie down devices shall be in accordance with MIL-STD-209, types one or three. If Air transportability is required (see 6.2) then tiedowns must be located and constructed so as to provide vehicle constraint aboard a C-5 aircraft in accordance with AFSC Design Handbook 1-11 and MIL-STD-1791.

3.14 Data plates. The following plates shall be installed in a conspicuous location on the end item.

3.14.1 Identification plate. A corrosion resistant metal name plate shall be permanently and legibly marked with the following information.

nomenclature
manufacturer's model number
manufacturer's serial number
special characteristics (HP, RPM, etc)
national stock number
manufacturer's name
date of manufacture
warranty expiration date
curb weight and CG location

3.14.2 Lubrication plates. A corrosion resistant metal lubrication plate(s) shall illustrate the lubrication points on the vehicle and shall identify the type of lubricant and service interval for each point.

3.14.3 Warning plates. Warning plates shall identify potentially hazardous conditions, locations and actions. Plates shall be sized and located so as to be conspicuous to the person at risk.

3.14.4 Instruction plates. All controls shall be labeled to identify the location and function of each control position. Instructions cast or stamped on or beside the controls are acceptable.

3.15 Treatment and painting. Mix trucks shall be painted in accordance with standard commercial practice provided that the paint and primer are both lead and chromate free. Color shall be 24052 of FED-STD-595 unless otherwise specified in 6.2. The exterior finish shall show no evidence of uneven application, runs, sags, curtains, orange peel, lack of adhesion, or other defect. All exterior trim items (such as wheels, wheel covers, running boards, metal radiator grilles, drip rails, bumpers, door handles, and light trim rings) shall be painted to match the chassis.

3.15.1 Markings. Markings (US Air Force and Registration Numbers) will be applied on dark green vehicles utilizing black paint color number 37038. Place registration numbers on each side of vehicle and on rear if space permits. Size of letters and numerals to be used are as follows:

US AIR FORCE 1.5 INCHES
REGISTRATION NUMBER 1.5 INCHES
REG NUMBER SPACING EXAMPLE 77D1977

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Lifting and tie down locations shall be identified by 1.5 inch letters stating "LIFT HERE" and "TIE DOWN HERE" as applicable.

3.15.2 Rustproofing. Unless otherwise specified (see 6.2), the vehicle shall be rustproofed in accordance with the contractor's standard commercial practice. The first article units shall not be rustproofed until after approval of the first article test report.

3.16 Workmanship. The workmanship shall be of the quality prevailing among manufacturers normally producing equipment of this type. Welded, bolted and riveted construction shall be in accordance with the highest industry standards.

4. QUALITY ASSURANCE.

4.1 Responsibility for inspection. The contractor is responsible for the performance of all inspection and testing requirements (henceforth, just testing) as specified herein. Except as otherwise specified, the contractor may use any inspection facilities acceptable to the government. Inspection records of the tests shall be kept complete and available to the government. The government reserves the right to perform any of the tests set forth in this specification where such tests are deemed necessary to assure that the vehicle conforms to the prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspection. Testing shall be of two types, First Article Testing and Acceptance Testing. First Article Testing shall be performed on a unit prior to acceptance of any unit by the government. The first article test sample may be a preproduction item or a production item which meets the requirements of this specification. The sample shall be identical to the items procured under the contract. Approval of the first article shall not constitute waiver by the government of its rights under the contract. First Article Testing shall consist of the requirements in section 4.4. Acceptance testing shall be performed on each unit prior to acceptance of that unit. Acceptance testing shall consist of the requirements in section 4.5.

4.3 Inspection conditions.

4.3.1 Apparatus. The apparatus used in conjunction with the testing specified herein shall be of laboratory precision and shall be calibrated at proper intervals to ensure accuracy.

4.3.2 Preparation for tests. The vehicle, equipped with all specified auxiliary equipment, shall be fully serviced and all equipment shall be prepared for immediate operation.

4.3.4 Test rejection criteria. Throughout all tests specified herein, the vehicle shall be closely observed for the following conditions, any of which shall be cause for rejection.

a. Failure to conform to design or performance requirements specified herein.
b. Spillage or leakage of any fluid, including fuel, coolant, lubrication or hydraulic fluid, under any operating condition except as allowed herein.
c. Structural failure, evidence of impending failure, or permanent deformation of any component or joint.
d. Evidence of excessive wear.

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e. Interference between vehicle components or between the vehicle and the ground (including all required obstacles).

f. Misalignment of any component.

g. Evidence of undesirable roadability characteristics, including instability in handling while cornering, braking, and traversing rough terrain as required.

h. Conditions which present a safety hazard to personnel during operation, service, or maintenance.

i. Overheating of the engine, transmission, or any other vehicle component.

4.4 Quality Conformance Inspection.

4.4.1 Examination. The First Article shall be examined for conformance to section 3.

TABLE 1

EXAMINATION

NO.	DESCRIPTION	PARA NO.
101	Commercial truck mixer as specified	3.1
102	Vehicle integrity as specified	3.1.2
103	Fluid filling, draining and checking	3.2.1
	Fill ports conspicuous and convenient	
	engine oil | | engine coolant | |	
	trans fluid | | hydraulic system(s) | |	
	Ability to check fluid level as specified	
	engine oil | | engine coolant | |	
	trans fluid | | hydraulic system(s) | |	
	Drain plugs as specified	
	engine oil | | engine coolant | |	
	trans fluid | | hydraulic system(s) | |	
104	Lube points as specified	3.2.2
105	DELETED	
106	General safety as specified	3.4
107	Grab rails, steps, platform, etc. as specified	3.4.1
108	Sharp edges treated as specified	3.4.4
109	Drum discharge opening as specified	3.6.1
110	Drum hatch as specified	3.6.1
111	Water system as specified	3.6.2
112	Discharge chute and extensions as specified	3.6.4
113	Hydraulic pump as specified	3.6.5
114	Hydraulic hoses as specified	3.6.5
115	Hydraulic filters as specified	3.6.5
116	Automatic oiler as specified	3.6.5
117	Mixer controls as specified	3.6.7
118	Fuel filters as specified	3.7.3
119	Cab as specified	3.10
120	Bumper and towing device as specified	3.11.1
121	Splash and stone throw protection as specified	3.11.3
122	Identification plate as specified	3.14.1
123	Lubrication plate as specified	3.14.2

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TABLE 1 (CONT)

EXAMINATION

NO.	DESCRIPTION	PARA NO.
124	Warning plate as specified	3.14.3
125	Instruction plate as specified	3.14.4
126	Paint results as specified	3.15.1
127	Markings as specified	3.15
128	Workmanship as specified	3.16

4.4.2 Preliminary operation. Start engine and turn on all lights. Lights shall remain on for the duration of the preliminary operation. Operate the empty vehicle in each gear in both high and low ranges for not less than one minute each. Horn, backup alarm, and all cab controls shall be operated sufficiently to verify proper function. Locking differentials, if unit is so equipped, shall be engaged and disengaged. If units are 6 X 6 and type B differentials are specified (see 6.2), front differentials shall be engaged and disengaged while the truck mixer is moving. The vehicle shall be stopped from 25 mph three times in succession using service brakes. Emergency brakes shall be used to stop the vehicle from 25 mph on a fourth occasion. The vehicle wheels shall be checked for overheating. All drum controls shall be operated sufficiently to determine proper function but not less than 20 times each. Vehicle shall be inspected for leaks and other rejection criteria listed in 4.3.4.

4.4.3 Performance testing. The following tests shall be performed consecutively.

a. Fill water tanks to capacity. Measure capacity.
b. Load drum with agregate. Agregate shall be of two distinct types, loaded separately. The proportions of each agregate shall be such that verification of homogenous mixing can be made.
c. The truck mixer shall be driven for 15 minutes over paved roads. The truck mixer shall travel at least 35 mph for five minutes and shall have the mixer operating these five minutes.
d. The truck mixer shall be driven at maximum safe speed over unimproved dirt roads for 60 minutes. Distance will vary depending on conditions but shall not be less than 6 miles.
e. The mixer shall be operated after the vehicle has stopped. One half the mixing rotations shall be controlled from the cab. A tachometer shall be used to check the governed engine speed. Stop the mixer and then complete mixing using the alternate control set.
f. Discharge one half the contents of the mixer. Take a sample from the first and fourth yard discharged. Stop drum and then complete discharge using the cab controls. Take a sample from the last yard discharged. Visually inspect the samples to ensure that the two aggregates are properly mixed.
g. Demonstrate chute maneuverability in accordance with the requirements of 3.6.4.
h. Demonstrate water injection by running at least 50 gallons into the drum. Wash off unit with wash off hose. Hose shall be of sufficient length to wash entire truck mixer without showering the washer.

4.4.4 Maintenance testing. After having performed the inspection according to 4.4.1, an authorized government representative shall select between four and all of the tasks listed in 3.2.4. These tasks shall be performed sequentially by one person in the sum of the time allotted for each task. All tools, parts, and supplies shall be on hand at the start of the test. The list of tasks performed and the total time taken to perform the tasks shall be recorded. The use of any tools other than common hand tools shall be noted. Power tools shall not be used.

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4.4.5 Lifting provisions test. Determine the load on each lifting attachment during static lift conditions. Apply a force of 2.3 times the respective static load to each of the lifting attachments and hold for 90 seconds. Inspect for deformation.

4.4.6 Tie down test. Each tiedown shall be tested to withstand its portion of the total loads as specified in MIL-STD-209. Inspect for deformation.

4.4.7 Final operation and inspection. The unit shall be examined for signs of any conditions for failure listed in 4.3.4.

4.4.8 Test report. The following shall be incorporated as a part of the test report:

a. All failures, maintenance performed and other events recorded shall be identified by accumulated operation time, miles, cycles or position of the test procedure, as appropriate. Test conditions at the time of failures or irregular operations identified shall be recorded.

b. Summary of the engineering reasoning and of any tests conducted to determine assignable causes for all failures and irregular operations identified.

c. Summary of the engineering reasoning behind any corrections made, to be made, or proposed to be made, and the predicted effectiveness of those corrections.

d. Test activity and/or contractor comments on item features or requirements, that if modified, should improve the item.

e. Test activity and/or contractor comments on field conditions or operating procedures to be avoided or cultivated to increase the reliability and useful life of the item.

f. The contractor shall furnish the following certifications. Each shall be an original signed by an authorized OEM representative and by an authorized contractor representative and shall clearly identify the type of item being certified, the limits to the certification, and the end item to which it will be installed.

(1) Certification that the truck mixer complies with 3.1.3 Operating conditions.
(2) Certification that noise level is as specified in 3.4.2 Noise Level.
(3) Certification that hydraulic hose complies with applicable requirements in 3.6.5 Drum Power Transmission.

g. The contractor shall furnish the following commercial spec sheets as part of the first article test report.

(1) Engine spec sheet.
(2) Transmission spec sheet.
(3) Chassis spec sheet.

4.5 Acceptance testing. Each production unit shall be tested in accordance with the following section.

4.5.1 Initial examination. Unit consists of identical components as the First Article. Unit is configured the same as the First Article. Workmanship is at least equal to that of the First Article.

4.5.2 Run-in testing. Run-in testing shall be performed according to the procedures in 4.4.2 Preliminary Operation.

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4.5.3 Final examination. Unit shall be inspected for the presence of any of the rejection criteria listed in 4.3.4.

4.6 Refurbishment. After successful completion of all testing and prior to delivery, each unit shall be refurbished to a like-new condition.

5. PACKAGING.

5.1 Packaging. Packaging shall be in accordance with the instructions of the procuring activity.

6. NOTES.

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This unit is intended for world-wide deployment. It will be used to mix, transport, and discharge concrete in support of both large and small construction projects. This unit is required to be capable of extended off road operation.

6.2 Ordering data.

a. Title, number, and date of this specification.
b. Name or number of any applicable amendments or supplement.
c. 6 X 4 or 6 X 6 required (see 1.2)
d. Winterization requirements, if other than as specified. State storage, operation, and starting extremes in degrees F. (see 3.1.3).
e. Type of differentials required (see 3.5.3).
f. Charging hopper, if different than standard (see 3.6.3).
g. Hydraulic circuit, if different than manufacturer's standard (see 3.6.5).
h. Air transportability, if required (see 3.13).
i. Paint color, if different (spec finish color and marking color per FED-STD-595) (see 3.15).
j. Rustproofing required, if other than the manufacturer's standard commercial practice (see 3.15.2).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerers whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirements for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Data requirements. The following Data Item Description (DIDs) must be listed, as applicable, on the Contract Data Requirements List (DD Form 1423) when this specification is applied on a contract, in order to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

DID Number	DID Title	Suggested Tailoring
DI-T-4901	FAT Procedures	See DD Form 1423
DI-T-4902	FA Inspection Report	See DD Form 1423
DI-T-2373A/T	Notification of Test	See DD Form 1423

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The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Forms 1423.

6.5 Subject term (key word) listing.

Chassis
Chute
Differentials
Drums
Hopper

Custodians:
Air Force - 99

Preparing activity:
Air Force - 84

Project No: 3895-F259

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-M-83805	30 MAR 90

3. DOCUMENT TITLE

Mixer, Concrete, Truck Mounted, 8 Cubic Yard Capacity, 6X4 or 6X6

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
WR-ALC/MMVRG Robins AFB, GA 31098-5609	(1) Commercial	(2) AUTOVON
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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